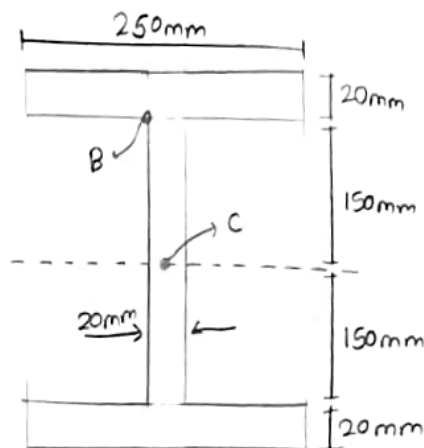
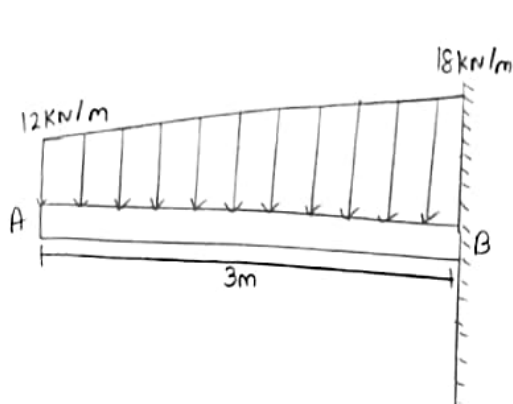


Name: Atharva S Bhosale

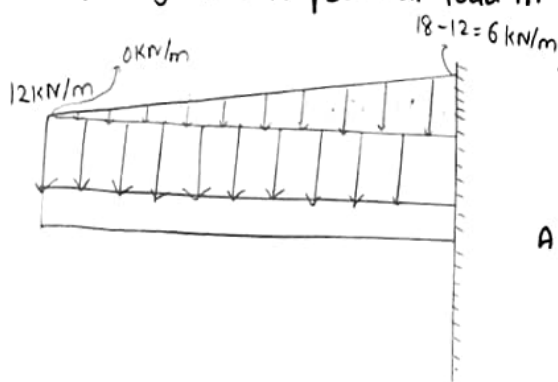
Roll No: ME248005

Q. The cantilever beam in the figure is made from a ductile material and is actually loaded. Determine the safety factor for the critical stress element using appropriate failure theory

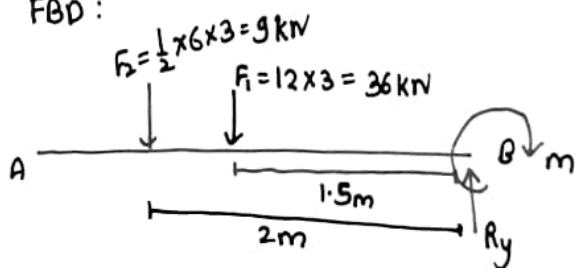
(Assume $\sigma_{yt} = 250 \text{ MPa}$)



Ans: Splitting the trapezoidal load in triangular load and rectangular load.



FBD:



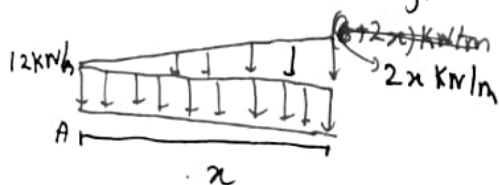
$$\sum M_B = 0 \Rightarrow m = F_1 \times 1.5 + F_2 \times 2 = 36 \times 1.5 + 9 \times 2$$

$$m = 75 \text{ kNm} \quad \boxed{m = 72 \times 10^6 \text{ Nmm}}$$

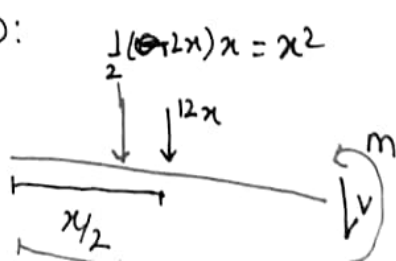
$$\sum F_y = 0 \Rightarrow R_y = F_1 + F_2 \Rightarrow R_y = 36 + 9 \Rightarrow \boxed{R_y = 45 \text{ kN}}$$

→ Shear force & Bending moment diagram.

Consider a element of length x from A



FBD:



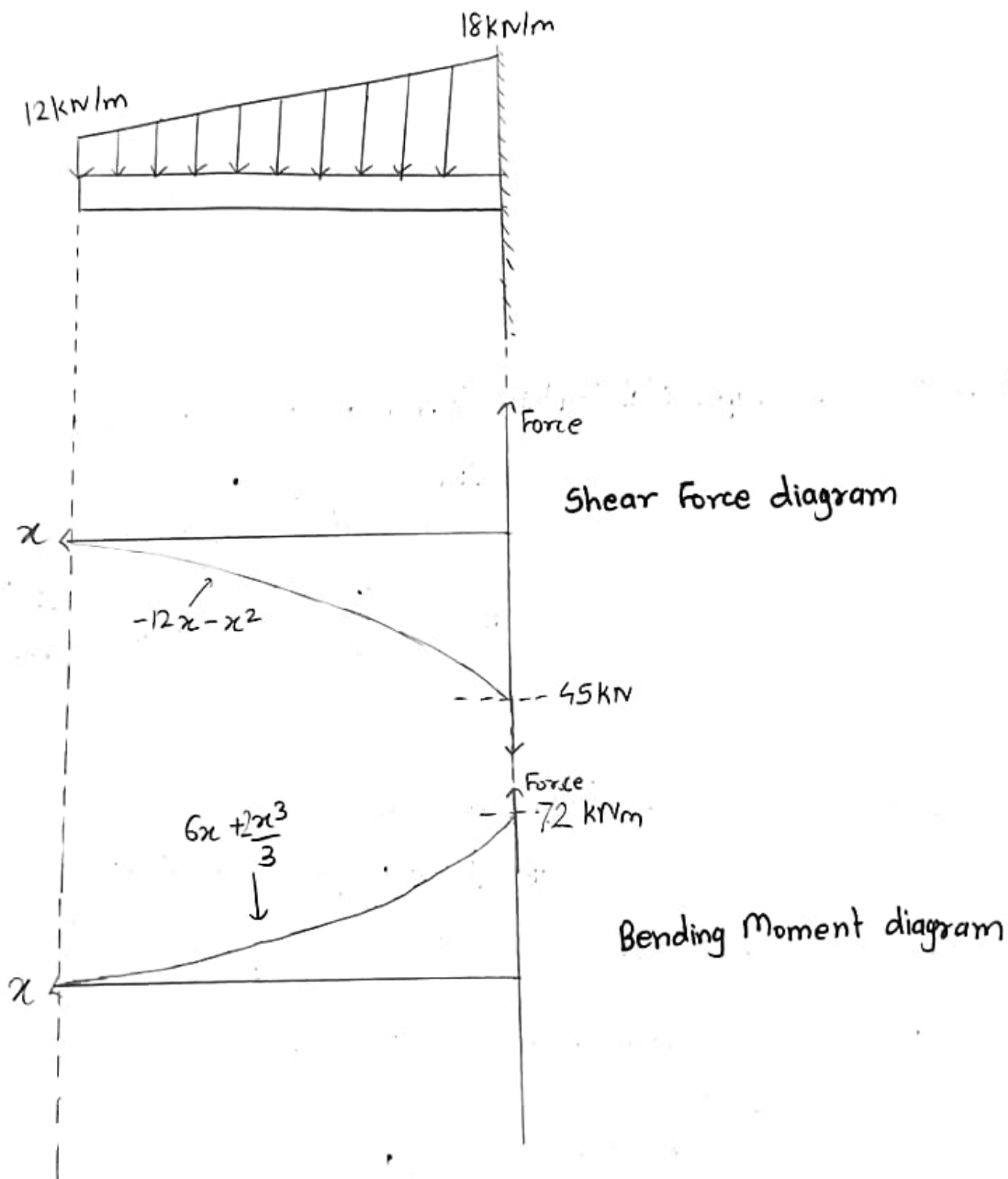
$$\sum F_y = 0 \Rightarrow V = -12x - \frac{1}{2}(6-2x)x$$

$$V = -12x - x^2$$

$$\sum M_A = 0$$

$$m - 12x \times \frac{x}{2} - x^2 \times \frac{x}{3} = 0$$

$$m = 6x^2 + \frac{x^3}{3}$$



Beam deflection;

$$EI \frac{d^2 y}{dx^2} = m(x)$$

where $y \rightarrow$ deflection about y axis.

$$EI \frac{d^2 y}{dx^2} = 6x^2 + \frac{2x^3}{3}$$

$$EI y' = 2x^3 + \frac{x^4}{6} + C_1$$

$$EI y = \frac{x^4}{2} + \frac{x^5}{30} + C_1 x + C_2$$

Boundary condition,

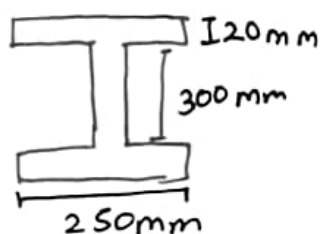
i) $x=3 \quad y=0$.

$\therefore C_2 = -48.6 - C_1$ (3)
 $C_2 = 153.9$

ii) $x=3 \quad y'=0$

$C_1 = -67.5$

Considering it as a structural steel $E=200 \text{ GPa}$.



Moment of Inertia about z will be.

$$I_z = \frac{1}{12} \times (300)^3 \times 20 + 2 \left(\frac{1}{12} \times 20^3 \times 250 + 20 \times 250 \times 160^2 \right)$$

$$I_z = 301.3 \times 10^6 \text{ mm}^4$$

Maximum deflection will occur at free end at pt A.
where $x=0$.

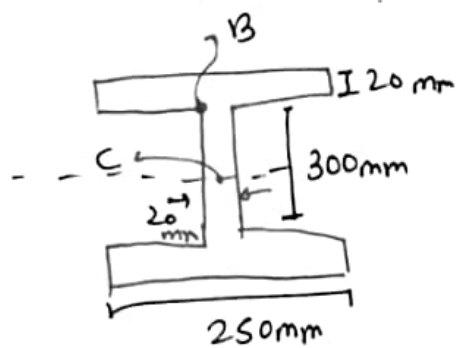
$$\therefore EI y(0) = C_2 = 153.9$$

$$y_{\max}^{(0)} = \frac{153.9 \text{ kNm}}{200 \times 10^9 \text{ Pa} \times 301.3 \times 10^6 \text{ mm}^4}$$

$$y_{\max}^{(0)} = 2.554 \text{ mm}$$

$\therefore$ maximum deflection will be 2.55 mm

Considering the flange,



At point B,

$$y = \frac{h}{2} = \frac{(300+40)}{2} = 170 \text{ mm} = 0.17 \text{ m}$$

Bending stress,

$$\sigma_B = \frac{M_{max} y}{I_z}$$

$$\sigma_B = \frac{72 \times 10^6 \times 170}{3.0133 \times 10^8} = 40.62 \text{ MPa}$$

Shear stress $\tau_B = 0$.

At point C,

$$y = 0$$

$$\text{Bending stress } \sigma_C = \frac{M y}{I} = 0$$

Shear stress,

$$Q = (250 \times 20) \times (150 + 10) + (150 \times 20) \times 75$$

$$Q = 1.025 \times 10^6 \text{ mm}^3$$

$$\tau_C = \frac{V_{max} Q}{I b} = \frac{45 \times 10^3 \times 1.025 \times 10^6}{3.0133 \times 10^8 \times 20} = 7.65 \text{ MPa}$$

$$\tau_C = 7.65 \text{ MPa}$$

Since stress at B > C critical point is B.

Principle stresses will be,

$$\sigma_1 = 40.62 \text{ MPa} \quad \sigma_2 = 0 \quad \sigma_3 = 0$$

And $\sigma_{yt} = 250 \text{ MPa}$ (given)

Using Von mises ,

$$\begin{aligned}\sigma_{vm} &= \sqrt{\sigma_1^2 - \sigma_1\sigma_2 + \sigma_2^2} \\ &= \sqrt{40.62^2 - 0 + 0}\end{aligned}$$

$$\sigma_{vm} = 40.62 \text{ MPa}$$

Safety factor,

$$n = \frac{\sigma_{yt}}{\sigma_{vm}} = \frac{250}{40.62} = 6.1546$$

$$\boxed{n = 6.1546}$$

Using Tresca theory,

$$\tau_{max} = \frac{\sigma_1 - \sigma_2}{2} = \frac{40.62 - 0}{2} = 20.31 \text{ MPa}$$

Safety factor

$$n = \frac{\sigma_{yt}}{2 \times \tau_{max}}$$

$$n = \frac{250}{2 \times 20.31} = 6.1546$$

$$\boxed{n = 6.1546}$$

	Quantity	Hand Calculation	Ansys	Difference (error)
1)	Max bending (von mises stress)	40.62 MPa	40.58 MPa	-0.1%
2)	Max shear stress (Tresca)	20.31 MPa	21.14 MPa	+4.1%
3)	Safety factor	6.15	6.16	+0.2%
4)	Max deformation	2.55 mm	2.42 mm	-5.4%